

① Define Angular and explain its primary purpose in web development.

→ Angular is a typescript based, open-source web Application.

→ it is developed by google.

→ it is build for single page web applications.

Purpose in web development:-

① component Based Architecture

② Two-way binding

③ dependency injection

④ Routing

⑤ Built in TOOLS and CLI

⑥ Form Handling

Angular is a robust front-end framework used to build modern, interactive & scalable web Applications.

② What is the primary difference between angular's two-way data binding & React's one-way dataflow?

Two-way-Binding

→ In angular two way binding the component and view are kept in sync.

→ When the user changes in the UI, the model is updated immediately.

Ex:- `<input [(ngModel)] = "username">`

React: One-way data flow

→ In React The data flows only from Component State to UI state.

→ When any changes occur you need to manually handle events.

Ex:- `<input value={username} onChange={() => submit setUser (e.target.value)}>`

Differences

Angular

- Two way direction
- UI updates Automatic
- Easier for small forms, inputs
- Less manual control

React

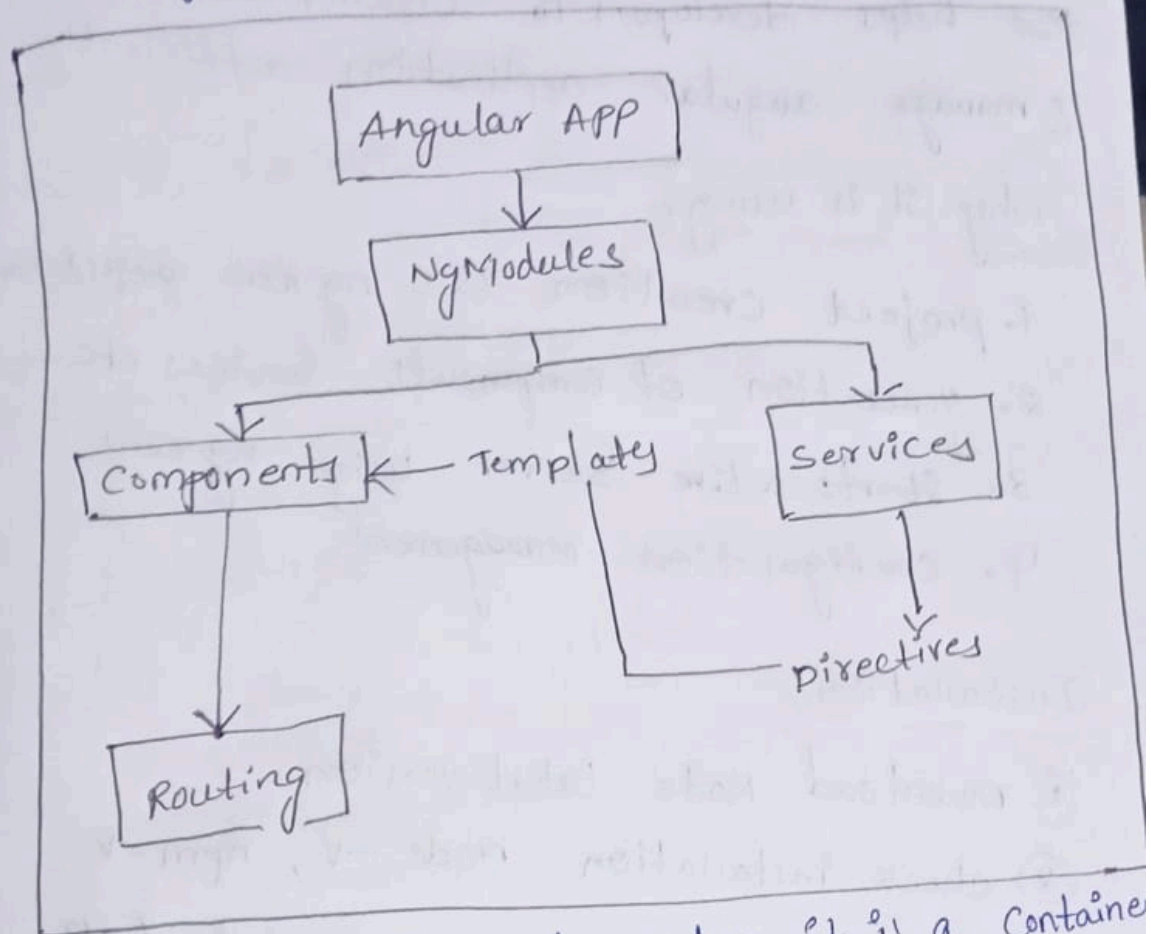
One way
manual

more predictable
controlled.

more explicit, better
for large-scale

③ Angular Architecture Diagram :

→ an Angular Application is made up of multiple Building Blocks to create interactive web pages.



NgModule : it is a decorator, it is a container for a group of related components, services --

- it imports the modules that are needed
- it exports the components to make available
- it provides services through the app module

④ Installing Angular CLI :

Angular CLI :

→ it's a powerful command line tool that helps developers to create, build, test & manage angular application efficiently.

Why it is using :

1. project creation ex: ng new project name
2. generation of components, services etc
3. starts a live Server using ng new
4. configuration management.

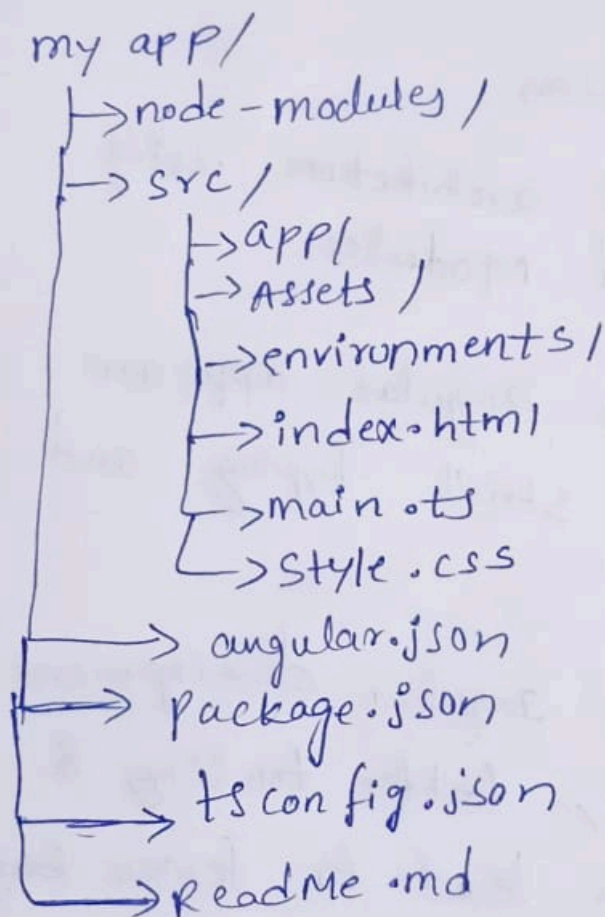
Installation :

- ① download Node latest version
- ② check installation node -v, npm -v
- ③ goto cmd ⇒ npm i -g @angular/cli
- ④ to confirm it is installed by type ng version

⑤ creation of Angular Application.

- To create a project
ng new myapp
- Type Y/N depending on whether you want or not features.
- cd myapp
- ng serve

default directories:-



⑥ Typescript in Angular :

Typescript is used in Angular because

- ① Helps to catch errors at compile-time
- ② Improved code quality
- ③ It supports oops concepts
- ④ Typescript compiles to plain Javascript, so it runs on any browser.

Typescript enhances the development process in Angular :

- ① Early error detection
- ② Encourages cleaner architecture using classes, interfaces and modules.
- ③ Scalability : Large angular apps are easier to manage with strict typing and structured code.
- ④ Typescript improves angular development by providing typesafety, better tooling & structured code which leads to fewer bugs.